

Swarup Kar Chaudhuri^[3pt]

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Professional Summary

Backend-focused software developer with a strong foundation in Python, REST APIs, FastAPI, PostgreSQL, and backend system development, currently specializing in Generative AI, Large Language Models, Retrieval-Augmented Generation (RAG), NLP, Deep Learning, and AI-powered applications. Experienced in building practical end-to-end projects involving document processing, vector databases, semantic search, embeddings, LLM integration, and intelligent backend services. Passionate about combining robust backend engineering with modern AI technologies to build reliable and useful production-oriented applications.

Skills

Technical Skills	Java, Python, SQL, FastAPI, REST APIs, React, PostgreSQL, SQLAlchemy, Generative AI, LLMs, RAG, NLP, Machine Learning, Deep Learning, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, Git, GitHub, Docker
Soft Skills	Problem Solving, Analytical Thinking, Communication, Teamwork, Adaptability, Ownership, Quick Learning, Continuous Learning

Projects

AI Document Assistant – Intelligent RAG Platform Python / FastAPI / React / PostgreSQL / ChromaDB

- Built a full-stack AI document assistant with FastAPI, React, PostgreSQL, and ChromaDB for intelligent document question answering.
- Developed an end-to-end **RAG pipeline** covering document parsing, chunking, embeddings, semantic retrieval, metadata, and context-aware generation.
- Integrated **Sentence Transformers** and **Groq Llama 3.3**; implemented secure REST APIs with JWT authentication and modular backend services.
- Added document upload, AI chat, conversation history, and responsive dark/light UI using React, TypeScript, and Tailwind CSS.

Enterprise AI Platform – Production-Oriented AI Workspace Python / FastAPI / React / PostgreSQL

- Built a production-oriented AI workspace for secure document management, semantic knowledge search, and document-aware AI conversations.
- Designed modular **FastAPI + React** architecture with REST APIs, authentication, persistent chat history, and user-specific workspace isolation.
- Implemented **RAG-based retrieval**, PostgreSQL/Alembic persistence, document metadata workflows, and extensible AI/agent components.
- Added backend unit/integration testing and Docker-ready deployment support with scalable service architecture.

Education

North Eastern Regional Institute of Science and Technology (NERIST)	2022–2026
Bachelor of Technology (B.Tech) – Electrical Engineering	

Achievements

- Built production-oriented AI applications using Python, FastAPI, RAG, ChromaDB, PostgreSQL, and LLM integrations.
- Solved **200+ algorithmic problems** on LeetCode and other competitive programming platforms.

Languages

English • Bengali • Hindi

Interests

Artificial Intelligence • NLP • RAG • LLMs • Gen AI • Backend Development